Number Sense
12625

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1. $622 - 66 =$ _____.
2. The product of the tens and thousands digits of 1254 is _____.
3. $432 + 978 =$ _____.
4. $7 + 8 \times 7 =$ _____.
5. $432 - 179 =$ _____.
6. The remainder of $243 \div 5$ is _____.
7. $15 \times 60 =$ _____.
8. $13 + 26 + 47 + 34 =$ _____.
9. The product of 17 and 4 is _____.
10. (estimate) $1423 - 345 + 1286 =$ _____.
11. $48 \times 11 =$ _____.
12. $849 - 534 - 213 =$ _____.
13. $22 \times 45 =$ _____.
14. The remainder of $110 \div 9$ is _____.
15. $17 \times 8 \times 5 =$ _____.
16. $17^2 =$ _____.
17. $9.6 \div 0.12 =$ _____.
18. *DCCCXLI* in Arabic numerals is _____.
19. $39 \times 41 =$ _____.
20. (estimate) $299 \times 411 =$ _____.
21. $25 \times 68 =$ _____.
22. $25^2 =$ _____.
23. The greater of $\frac{4}{7}$ and $\frac{7}{12}$ is _____ (fraction).
24. If 1 stone is equal to 14 pounds, then 98 pounds is equal to _____ stones.
25. $6 \times 57 \div 38 =$ _____.
26. The GCD of 20 and 26 is _____.
27. $714 \div 17 =$ _____.
28. $\frac{1}{4} + \frac{5}{12} =$ _____ (fraction).
29. $33 + 37 + 41 + 45 + 49 =$ _____.
30. (estimate) $44 \times 58 \times 62 =$ _____.
31. $73 \times 99 =$ _____.
32. The LCM of 20 and 26 is _____.
33. $85^2 =$ _____.
34. $1200 \times 10^{-2} =$ _____.
35. The area of a rectangle with perimeter 24 and width 3 is _____.
36. 60% of 80 is _____.
37. 3 hours is _____ seconds.
38. The number of multiples of 9 between 25 and 88 is _____.
39. $71 \times 79 =$ _____.
40. (estimate) $635241 \div 126 =$ _____.

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41. $4^4 =$ _____.

42. The remainder of $8947 \div 11$ is _____.

43. The twenty-second term of the arithmetic sequence $18, 36, 54, \ldots$ is _____.

44. $101 \times 56 =$ _____.

45. The sum of the measures of the interior angles in a regular hexagon is _____ °.

46. $39 \times 79 =$ _____.

47. The median of the list $18, 13, 5, 11$ is _____.

48. $14 \times 13 + 26 \times 3 =$ _____.

49. $54^2 - 34^2 =$ _____.

50. (estimate) $571428 \times 8 =$ _____.

51. $105 \times 116 =$ _____.

52. $63^2 =$ _____.

53. $9\frac{1}{11}\%$ of 99 is _____.

54. The sum of the terms of the arithmetic sequence $7, 12, 17, \ldots, 57$ is _____.

55. $71 \times 111 =$ _____.

56. The length of the hypotenuse of a right triangle with legs of length 7 and 24 is _____.

57. 1100110_2 in base 8 is _____ $_8$.

58. The diameter of a semicircle with perimeter $16 + 8\pi$ is _____.

59. $83 \times 97 =$ _____.

60. (estimate) $54 \times 5556 =$ _____.

61. The number 78 in base 7 is _____ $_7$.

62. Two standard six-sided dice are rolled. The probability the sum of the numbers shown is 4 is _____ (fraction).

63. $0.3\overline{4} =$ _____ (fraction).

64. $9\frac{2}{5} \times 9\frac{3}{5} =$ _____ (mixed number).

65. The area of an isosceles right triangle with a hypotenuse of length 14 is _____.

66. $125 \times 721 =$ _____.

67. If $x = 24$, then $x^2 - 8x =$ _____.

68. $9 \times 13 \times 12 \times 8 =$ _____.

69. $34^2 + 37^2 =$ _____.

70. (estimate) $5.1^4 =$ _____.

71. $16^{\frac{7}{4}} =$ _____.

72. The number of positive whole number divisors of 63 is _____.

73. $314 \times 999 =$ _____.

74. The surface area of a cube with edge length 8 is _____.

75. $\sqrt{72} \times \sqrt{200} =$ _____.

76. $997^2 =$ _____.

77. $1007 \times 1013 =$ _____.

78. The sum of the terms of the geometric sequence $\frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots$ is _____.

79. $121 \times 34 \times 35 =$ _____.

80. (estimate) $\sqrt[3]{2026 \times 4052} =$ _____.

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1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. What is $5 + 10 + 16 + 20 + 26 + 30 + 36$?

(A) 133 (B) 143 (C) 140 (D) 138 (E) 113

2. Two standard six-sided dice are rolled. What is the probability that the numbers rolled are 1 apart?

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $\frac{7}{36}$ (D) $\frac{5}{36}$ (E) $\frac{5}{18}$

3. What is the sum of all positive whole numbers less than 200 that are multiples of 4 but not 8?

(A) 2500 (B) 1250 (C) 1000 (D) 2000 (E) 1600

4. Avery is building a tower with n floors. The number of dollars that each floor costs is equal to 24 times the square of the number of the floor. For example, the second floor costs 96 dollars to build, and the fourth floor costs 384 dollars to build. Given that Avery has a total of 1944 dollars to build the tower, what is the greatest possible value of n?

(A) 7 (B) 8 (C) 9 (D) 5 (E) 6

5. In how many ways can the letters in the word *FAMILY* be arranged such that exactly two but not all of $\{A, I, Y\}$ are consecutive?

(A) 288 (B) 324 (C) 144 (D) 576 (E) 432

6. The coordinates of points A and B in the xy-plane are $(0, 2)$ and $(2, 0)$, in that order. What is the sum of the coordinates of the midpoint of segment $\overline{AB}$?

(A) -1 (B) 4 (C) 2 (D) 0 (E) 1

7. How many three-digit positive whole numbers are divisible by 12, 16, or both?

(A) 117 (B) 120 (C) 110 (D) 113 (E) 112

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8. Henry rolls three standard 6-sided dice, one at a time. What is the probability that the sum of the numbers rolled on the second and third dice is a multiple of the number rolled on the first die?

(A) $\frac{11}{27}$ (B) $\frac{5}{12}$ (C) $\frac{29}{72}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

9. Emily and Fred are picking berries, and they each pick a whole number of berries. In addition, the total number of berries that they picked is equal to 8 times the positive difference between the numbers of berries that the two of them picked. Given that Fred picked a perfect-square number of berries and that Emily picked more berries than Fred, what is the least possible number of berries that Emily could have picked?

(A) 18 (B) 63 (C) 9 (D) 40 (E) 56

10. A regular pentagon, a square, and a third regular polygon all share a corner, and they do not overlap. Each one of these three polygons also shares a single side with each of the two other polygons. How many sides does the third polygon have?

(A) 15 (B) 10 (C) 12 (D) 24 (E) 20

11. A specific brand of lemonade is 10 percent heavier than water of the same volume. Brady decides that this lemonade is not sweet enough for him, so he adds 10 grams of sugar to 220 grams of lemonade. Given that the sugar does not increase the lemonade's volume, by what percent is Brady's lemonade heavier than water of equal volume?

(A) 17 (B) 15 (C) 20 (D) 25 (E) 22.5

12. Turner has a triangle with an area of 72 and a base of length 16. On the other hand, Leo has a triangle with an area of 98 and a base of length 7. What is the sum of the heights of these two triangles?

(A) 37 (B) $\frac{37}{2}$ (C) 147 (D) 35 (E) 33

13. Patricia has a bag of 48 marbles, some of which are blue and the rest of which are green. One day, Patricia puts 24 more blue marbles in the bag, which doubles her chance of picking a blue marble when randomly picking a single marble from the bag. Given that Patricia started with at least one marble of each color, how many blue marbles did Patricia start with?

(A) 48 (B) 30 (C) 18 (D) 12 (E) 24

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14. Call a positive whole number *fortunate* if it contains at least three consecutive 7s. How many positive whole numbers less than 100000 are fortunate?

(A) 299 (B) 300 (C) 280 (D) 269 (E) 289

15. Three congruent regular hexagons with side length 3 are joined at a single vertex so that they share one side with each of the other hexagons but do not overlap. What is the perimeter of the smallest equilateral triangle that can completely enclose the resulting 12-sided shape?

(A) $18\sqrt{3}$ (B) 27 (C) 18 (D) $9\sqrt{3}$ (E) $27\sqrt{3}$

16. Oliver mows his lawn on Monday, Wednesday, and Friday every week. On each day that Oliver mows his lawn, he can mow either 10% of the total area of his lawn or 20% of the remaining unmowed area of the lawn. In addition, Oliver will always mow the entire lawn before mowing any part of it twice. Given that he starts mowing on a Wednesday, at least how many days are between the day he starts and the day on which he finally mows the lawn completely for the first time?

(A) 21 (B) 19 (C) 17 (D) 16 (E) 14

17. Jimmy's drawer has 25 different pieces of clothing, each of which is either a shirt or a pair of shorts. What is the greatest possible number of ways for Jimmy to pick one shirt and one pair of shorts to wear?

(A) 156 (B) 150 (C) 625 (D) 225 (E) 50

18. Triangle ABC has side lengths $AB = 13$, $BC = 14$, and $CA = 15$. Points P and Q lie on $\overline{BC}$ and $\overline{CA}$, in that order, such that $PC = 3$, $PQ = 4$, and $CQ = 5$. What is the area of $\triangle ABQ$?

(A) 72 (B) 78 (C) 56 (D) 460 (E) 64

19. Juan selects three of the twelve months of 2026 and puts them in chronological order, with January considered as the first month. There cannot be more than three months strictly between the first and second months chosen or the second and third months chosen. For instance, Juan could select January, May, and August. Given that no two chosen months can be consecutive, how many unique combinations of months can Juan select?

(A) 84 (B) 24 (C) 54 (D) 51 (E) 48

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20. The product of the positive whole numbers k and $2k-1$ has a units digit of 8. What is the greatest possible value of k less than 100?

(A) 92 (B) 96 (C) 98 (D) 90 (E) 94

21. A *lattice point* is a point with integer coordinates. For example, $(1,2)$, $(-3,7)$, and $(-4,0)$ are all lattice points, but $(-\frac{1}{2},99)$ is not. How many lattice points lie strictly inside the graphs of both $x^2+3y^2=9$ and the square with vertices at $(\pm 2,\pm 2)$ in the xy-plane?

(A) 12 (B) 13 (C) 11 (D) 9 (E) 15

22. Billy invents a new math symbol $\#$ that is defined as $a\#b=a^2+b$. What is the value of $8\#3$?

(A) 73 (B) 67 (C) 64 (D) 17 (E) 14

23. At a party, there are 10 people and 10 different coats, with each coat belonging to exactly one person. After the party, each person randomly selects a coat to take home. What is the expected number of people who select their own coat?

(A) 0 (B) 2 (C) 3 (D) 4 (E) 1

24. Parallel lines ℓ and m both intersect perpendicular lines a and b. In addition, both a and b intersect ℓ at point P, a intersects m at point Q, and b intersects m at point R. Given that $QR=75$ and $PQ=21$, what is the area of triangle PQR?

(A) 750 (B) 756 (C) 720 (D) 504 (E) 525

25. The square of the difference between the digits of each two-digit positive whole number is calculated. What is the sum of all of these squares?

(A) 1178 (B) 1322 (C) 1365 (D) 910 (E) 1161

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26. An arithmetic sequence and a geometric sequence share the same first and third terms. Given that the common difference of the arithmetic series is 26 and that the third term is 65, what is the fifth term of the geometric sequence?

(A) 169 (B) 300 (C) 260 (D) 234 (E) 325

27. While ordering his meal at a restaurant, Hungry Harry can select up to 2 of 4 different appetizers, up to 2 of 7 different entrees, and up to 2 of 5 different desserts. Harry will select at least one of each. In addition, he will always select exactly 1 entree if he orders exactly 2 appetizers, and he will always select exactly 2 desserts if he orders exactly 1 entree. In how many ways can Harry order his meal?

(A) 1900 (B) 2040 (C) 2026 (D) 1960 (E) 2000

28. In a circle with radius 2, a sector with a central angle is n degrees is drawn. Given that the area of the sector is numerically equal to the perimeter of the rest of the circle, including the radii from the sector, the value of n can be expressed in the form $\frac{a}{\pi} + b$ for positive whole numbers a and b. What is $a + b$?

(A) 360 (B) 270 (C) 180 (D) 225 (E) 315

29. For how many positive whole numbers n less than 1000 is $n^2 + n$ a multiple of 12?

(A) 333 (B) 332 (C) 665 (D) 664 (E) 330

30. The graph of a quadratic polynomial with leading coefficient 1 passes through the points $(0, 50)$, $(m, 20)$, and $(2m, 50)$ in the xy-plane for some real number m. What is m^2?

(A) 20 (B) 120 (C) 15 (D) 30 (E) 50

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Problems 1 & 2

1. In an equilateral triangle, a segment is drawn from each corner of the triangle to the midpoint of the opposite side. Including the original equilateral triangle, how many triangles are formed by the three original sides and the three new segments?

1.

2. What is the sum of all positive prime numbers less than 100 with the property that they can be expressed as the sum of two other positive prime numbers?

2.

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Problems 3 & 4

3. Helen has a whole number of brownies that can be split evenly into 7 groups. However, a bandit steals some of the brownies, and she realizes that she can now split the brownies evenly into 13 groups with 40 percent as many brownies in each group. What is the least possible whole number of brownies that were stolen?

3.

4. John randomly picks three not necessarily distinct whole numbers between 1 and 100, inclusive. What is the probability that these three numbers can represent the degree measures of the angles of an isosceles triangle? Express your answer as a common fraction.

4.

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Problems 5 & 6

5. A chessboard can be represented as an 8×8 grid of unit squares. When a rook is placed on a unit square on the board, it attacks anything that is in the same row or column as itself. At most how many rooks can be placed on the chessboard so that none of them attack any of the other rooks?

5.

6. When the expression $(x+2)^7$ is expanded, what is the coefficient of the x^3 term?

6.

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Problems 7 & 8

7. A box contains 3 red marbles, 4 green marbles, and 5 yellow marbles. Two marbles are drawn at random without replacement. What is the probability that the two marbles are the same color given that at least one of them is yellow? Express your answer as a common fraction.

7.

8. Triangle ABC has side lengths $AB = AC = 15$ and $BC = 24$, and it is inscribed in circle O. Triangle ADE is also inscribed in circle O such that $\overline{DE}$ is a diameter of O and $\overline{DE} \parallel \overline{BC}$. What is the ratio of the area of $\triangle ADE$ to the area of $\triangle ABC$? Express your answer as a common fraction.

8.

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Team Round
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6.	7.	8.	9.	10.

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1. A snack shop sells 12 hot dogs, 8 cans of soda, 15 bags of popcorn, and 6 pieces of candy during a game. Given that hot dogs cost 3 dollars each, cans of soda cost 2 dollars each, bags of popcorn cost 4 dollars each, and pieces of candy cost 1 dollar each, how many dollars did the snack shop earn?

2. Gus orders exactly one ice cream flavor per day. Each day, there is a 40 percent chance that he orders vanilla, a 50 percent chance that he orders chocolate, and a 10 percent chance that he orders strawberry. What is the probability that, over a stretch of four consecutive days, Gus orders vanilla on the first day, chocolate on the last day, and never chooses the same flavor on any two consecutive days? Express your answer as a common fraction.

3. A theater sells three types of tickets. For every 5 balcony tickets sold, there are 3 mezzanine tickets sold. For every 4 mezzanine tickets sold, there are 9 orchestra tickets sold. In addition, balcony, mezzanine, and orchestra tickets cost 15, 25, and 40 dollars each, in that order. Given that the total cost of all of the tickets sold was 11760 dollars, how many orchestra tickets were sold?

4. Triangle ABC has whole-number side lengths and a right angle at vertex B. A circle with radius 5 centered at A intersects sides $\overline{AB}$ and $\overline{AC}$ at points D and E, in that order. Given that $CE = 4 \cdot BD$, what is the least possible area of triangle ABC?

5. How many two-digit odd positive whole numbers are divisible by the product of their digits?

6. What is the sum of the least and greatest positive values of x at which the value of the function $f(x) = x - 10\sqrt{x} + 21$ is not positive?

7. How many right triangles with whole-number side lengths have 60 as their longer leg length?

8. In the prime factorization of 50!, what is the sum of all of the exponents of the prime factors?

9. A regular octagon can be inscribed in a circle with radius 9. A second circle is drawn so that the center of the octagon as well as two of its vertices lie on it. Given that those two vertices are not on the same side of the octagon, the least possible area of the second circle can be expressed in the form $\frac{a}{b}\pi$ for relatively prime positive whole numbers a and b. What is $a + b$?

10. The square of the positive real solution to the equation $\frac{x^2+3}{x^2+4} - \frac{x^2+1}{x^2+2} = \frac{1}{2026}$ can be expressed in the form $\sqrt{a} - \sqrt{b}$ for positive whole numbers a and b. What is $a + b$?

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